KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Computer Science and Engineering
Spring-2022 Final Examination
Program: B.Sc. in CSE (Batch – 13th) 1st Year 1st Semester
Course Title: Physics - I
Course Code: PHY 1101

PART-A: MCQ

Time: 10 Minutes **Total Marks: 10x1=10**

Student's ID:

Date: **Invigilator's Signature:**

(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

1. What is the value of proportionality constant of Biot-Savart Law?

   A) $4 \times 10^{-7} TmA^{-1}$

   B) $4\pi \times 10^{-7} TmA^{-1}$

   C) $2 \times 10^{-7} TmA^{-1}$

   D) $2\pi \times 10^{-7} TmA^{-1}$

   E) $10^{-7} TmA^{-1}$

2. If two bodies are connected by a massless spring to create a two-body system and one of the bodies having mass of $4g$, what is the possible amount of reduced mass of that system?

   A) $10g$

   B) $8g$

   C) $6g$

   D) $4g$

   E) $2g$

3. Mutual induction occurs in the ____________ .

   A) magnet

   B) RC circuit

   C) transformer

   D) battery

   E) resistance

4. What is the unit of the e.m.f.?

   A) $Volt$

   B) wb

   C) $Gauss$

   D) T

   E) A

5. The maximum displacement of a simple pendulum is called the …………. of the pendulum.

   A) frequency

   B) time period

   C) velocity

   D) acceleration

   E) amplitude

6. A force of 4N on the spring produces a displacement of 0.02m. What is the force constant of the spring?

A) $400Nm^{-1}$

B) $200Nm^{-1}$

C) $100Nm^{-1}$

D) $0.08Nm^{-1}$

E) $0.05Nm^{-1}$

7. Frequency of the electric supply line for home in Bangladesh ______ .

A) $10Hz$

B) $50Hz$

C) $100Hz$

D) $220Hz$

E) $311\mathrm{Hz}$

8. What will happen when the n-pole of the magnet moves towards a coil?

A) Clockwise current induced in the coil.

B) Anti-clockwise current induced in the coil.

C) Attract each other.

D) No change of magnetic flux in the coil.

E) Demagnetize the magnet.

9. The ratio between rms & average value of AC is ___________ .

A) 0.01

B) 0.10

C) 0.11

D) 1.11

E) 11.1

10. If the closed path does not link with the current loop, then the Ampere's law –

A) $\oint_l \bar{B}.\overline{dl} = 0$

B) $\oint_l \bar{B}.\overline{dl} = \mu_0$

C) $\oint_l \bar{B}.\overline{dl} = i$

D) $\oint_l \bar{B}.\overline{dl} = n$

E) $\oint_l \bar{B}.\overline{dl} = BiL$

Signature of Examiner

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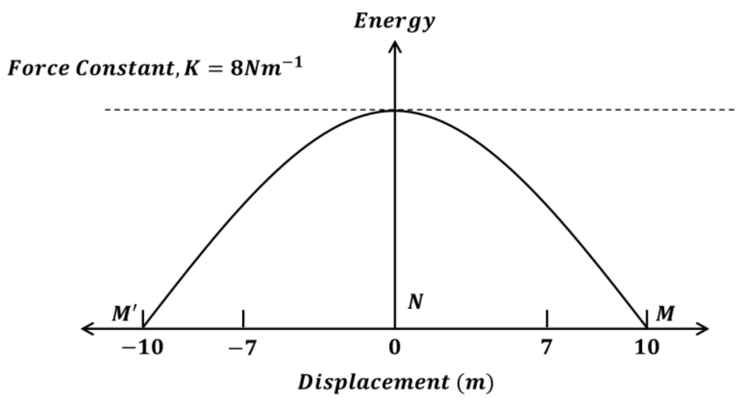
PART-B: SAQ

Time: 1 Hour and 50 Minutes **Total Marks: 5x6=30**

INSTRUCTIONS: Answer question no. 01 (one) and any 04 (four) from the rest. Marks are shown by side.

1. See the energy distribution curve of a spring pendulum and answer the questions that follows:



a) What is the difference between oscillation and rotational motion? [1]

b) Narrate the velocity and acceleration at point N. [1]

c) For which energy does this curve apply? – Justify your answer for M, N & M' points. [4]

2. a) The differential equation of a SHO is $\frac{d^2x}{dt^2} + 81x = 0$; if the mass of the oscillator is 20gm then find the value of angular velocity & force constant. [3]

b) If two bodies of masses $5g$ and $10g$ are connected by a massless spring of force constant 1.5 dynes/cm, then calculate the effective mass and time period of oscillation. [3]

3. a) State the Biot-Sevart law. What is the vector form of it? [2]

b) Two long straight conductors are held parallel to each other at a distance 0.3m. If the conductors carry currents 2A and 8A in opposite directions, then what is the distance of the neutral point from the conductor carrying smaller current? [4]

4. a) Write some properties of solenoid. [1]

b) Explain the force between two parallel current carrying conductors. [3]

c) Distinguish step up and step-down transformer. [2]

5. a) Define 1 Henry. [1]

b) Two nearby coil A and B have number of turns respectively of 200 and 1000. When $2A$ current flows through the coil, then magnetic flux $2.4 \times 10^{-4} wb$ is produced in the coil A and $1.6 \times 10^{-4} wb$ in the coil B. Calculate (i) the self – inductance of the coil A (ii) the mutual inductance of the coil B. (iii) If current stops A in $0.4\ sec$, find the inducted emf in coil B. [4]

6. a) Illustrate AC & DC graphically with proper indication. [2]

b) Investigate the following quantities from alternating current wave equation

$$i = 50\ sin\ 628t$$

(i) Peak value; (ii) Frequency; (iii) RMS value; (iv) Average value of the current. [4]